Data Science

Prof. Rayson Laroca
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About me

- Rayson Laroca

- Researcher with PPGIa/PUCPR
  - https://www.ppgia.pucpr.br/~rayson/
  - Research topics:
    - Computer Vision
    - Deep Learning
    - Pattern Recognition

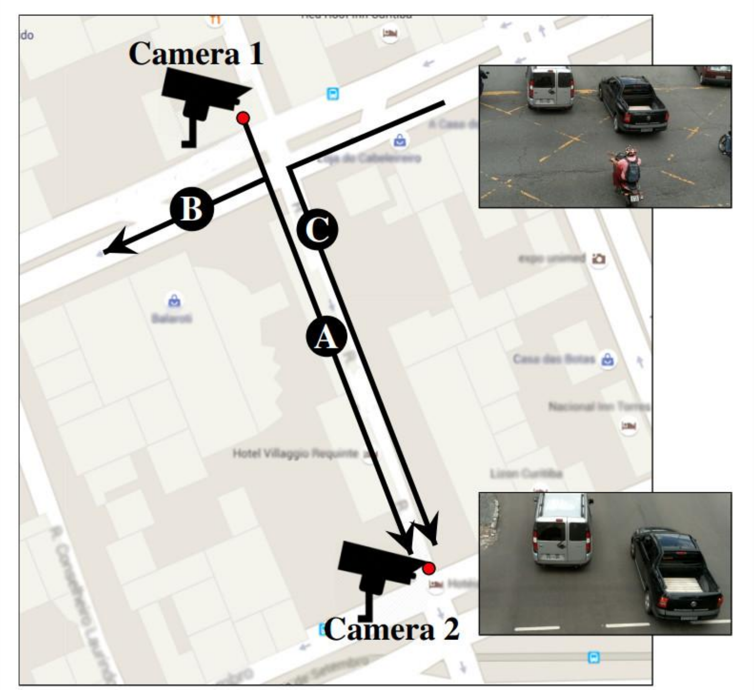
- Contact via Canvas

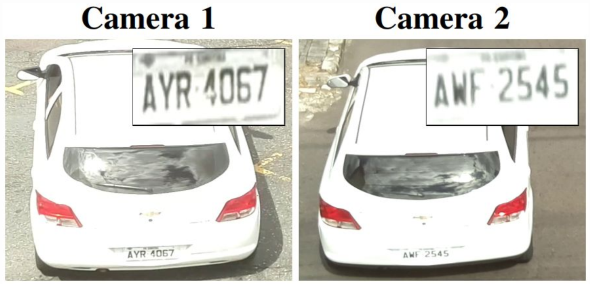
About me

- Education:
    - B.Sc. in Software Engineering (UEPG)
        - Exchange at the University of Coimbra (UC), Portugal
    - M.Sc. and Ph.D. in Computer Science (UFPR)
- Currently conducting postdoctoral research at UFPR

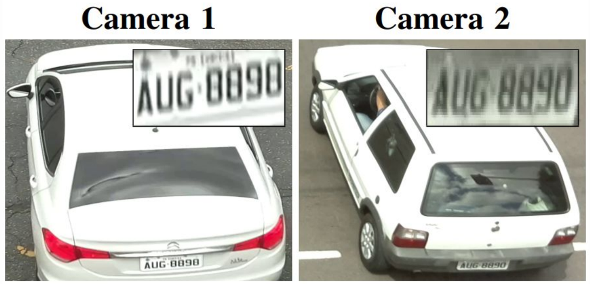
Research Overview

Vehicle Identification





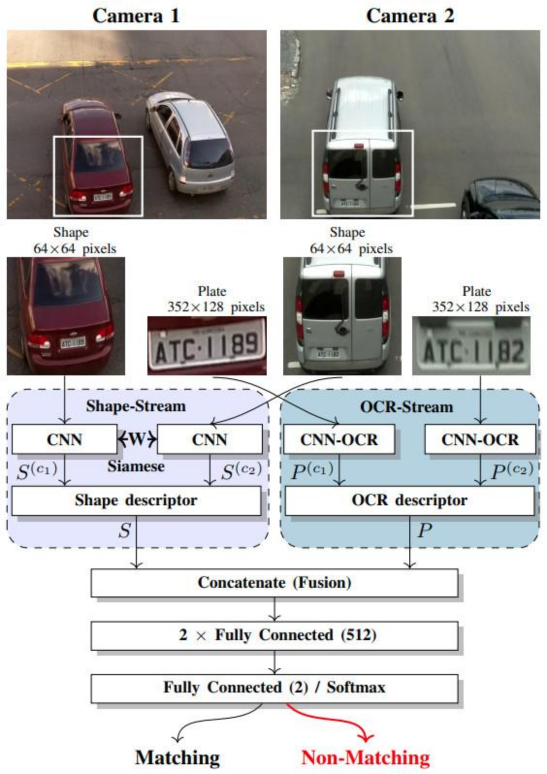
(a) Similar vehicles with dissimilar LPs.



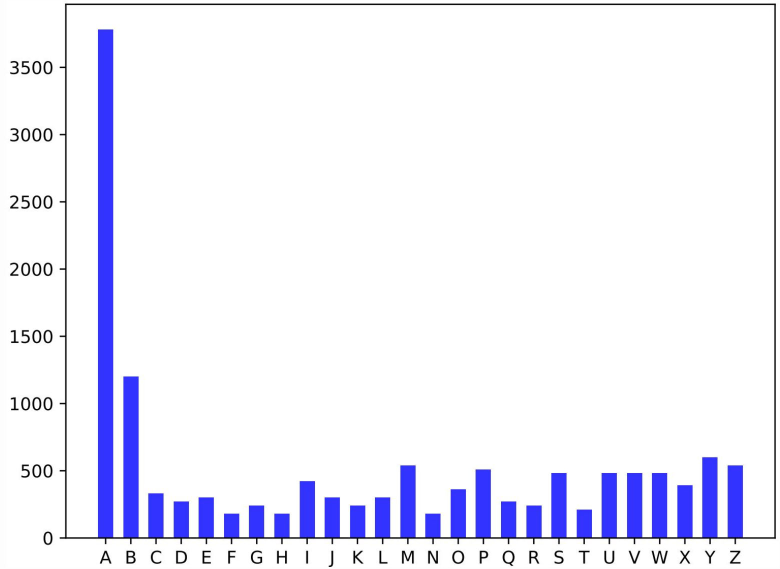
(b) Dissimilar vehicles with similar LPs.

I. O. de Oliveira, R. Laroca, D. Menotti, K. V. O. Fonseca, R. Minetto, "*Vehicle-Rear: A New Dataset to Explore Feature Fusion for Vehicle Identification Using Convolutional Neural Networks,*" IEEE Access, vol. 9, pp. 101065-101077, 2021.

Vehicle Identification



Class imbalance



R. Laroca, E. Severo, L. A. Zanlorensi, L. S. Oliveira, G. R. Gonçalves, W. R. Schwartz, D. Menotti, "*A Robust Real-Time Automatic License Plate Recognition Based on the YOLO Detector*," in International Joint Conference on Neural Networks (IJCNN), 2018.

Vehicle Identification

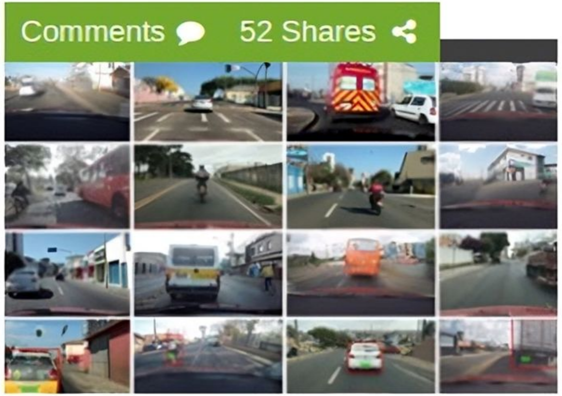
Synthetic Data

R. Laroca et al., "*Advancing Multinational License Plate Recognition Through Synthetic and Real Data Fusion: A Comprehensive Evaluation,*" IET Intelligent Transport Systems, 2025.

Vehicle Identification

NVIDIA DEVELOPER

NEWS CENTER



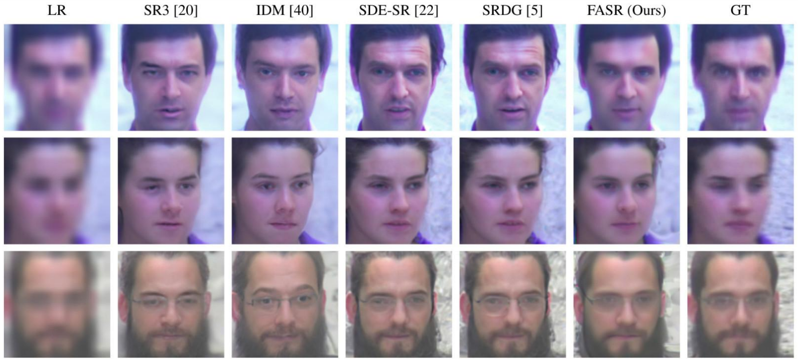
Researchers Develop AI System for License Plate Recognition

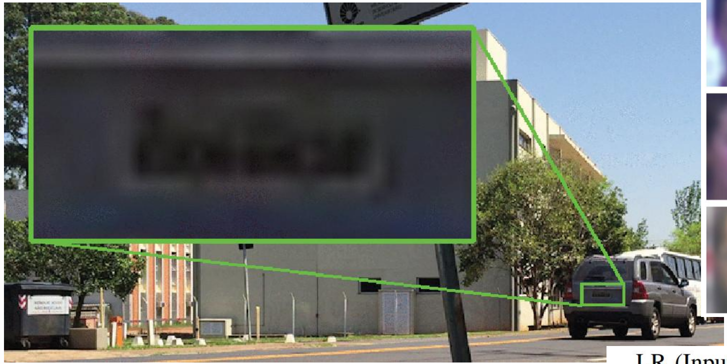
April 5

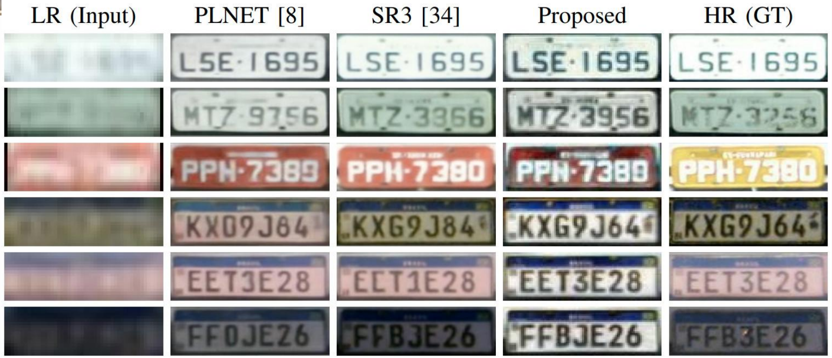
Researchers in Brazil developed a deep learning system that captures license plate data in real-time, resulting in better performance than most commercially available products in Brazil.

Super-Resolution

M. dos Santos, R. Laroca, R. O. Ribeiro, J. C. Neves, D. Menotti, "*Multi-Feature Aggregation in Diffusion Models for Enhanced Face Super-Resolution,*" in Conference on Graphics, Patterns and Images (SIBGRAPI), Sept. 2024.







V. Nascimento, R. Laroca, R. O. Ribeiro, W. R. Schwartz, D. Menotti, "*Enhancing License Plate Super-Resolution: A Layout-Aware and Character-Driven Approach,*" in Conference on Graphics, Patterns and Images (SIBGRAPI), 2024.

Wagon Counting and Identification

R. Laroca, A. C. Boslooper, D. Menotti, "*Visual Analytics Keep Track of Freight Wagons*," Railway Gazette International, vol. 176, no. 10, pp. 38-39, 2020.

Level Crossing Control and Monitoring

Software currently licensed to a company in the railway sector.

Drone Detection

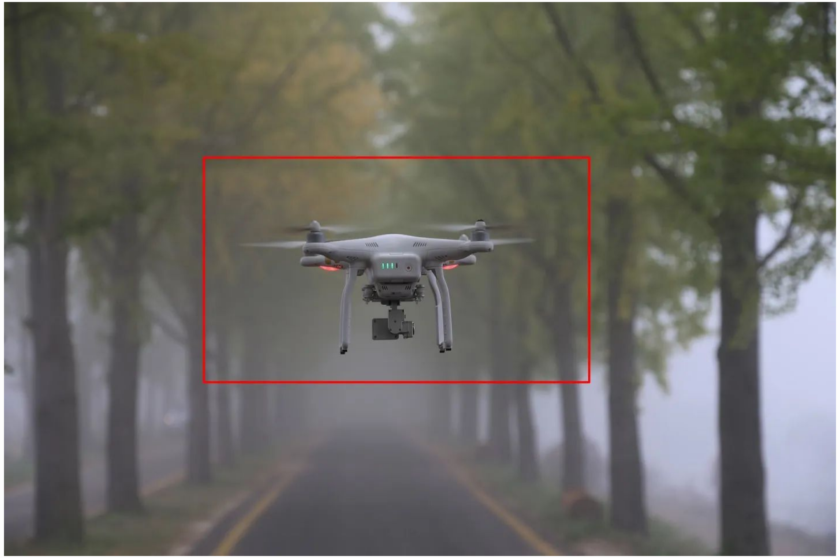
8th WOSDETC Drone-vs-Bird Detection
Data Competition @IJCNN25

UNIVERSITÀ DEL SALENTO
L'Ateneo tra i due mari

Fraunhofer IOSB

CERTH
CENTRE FOR RESEARCH & TECHNOLOGY HELLAS

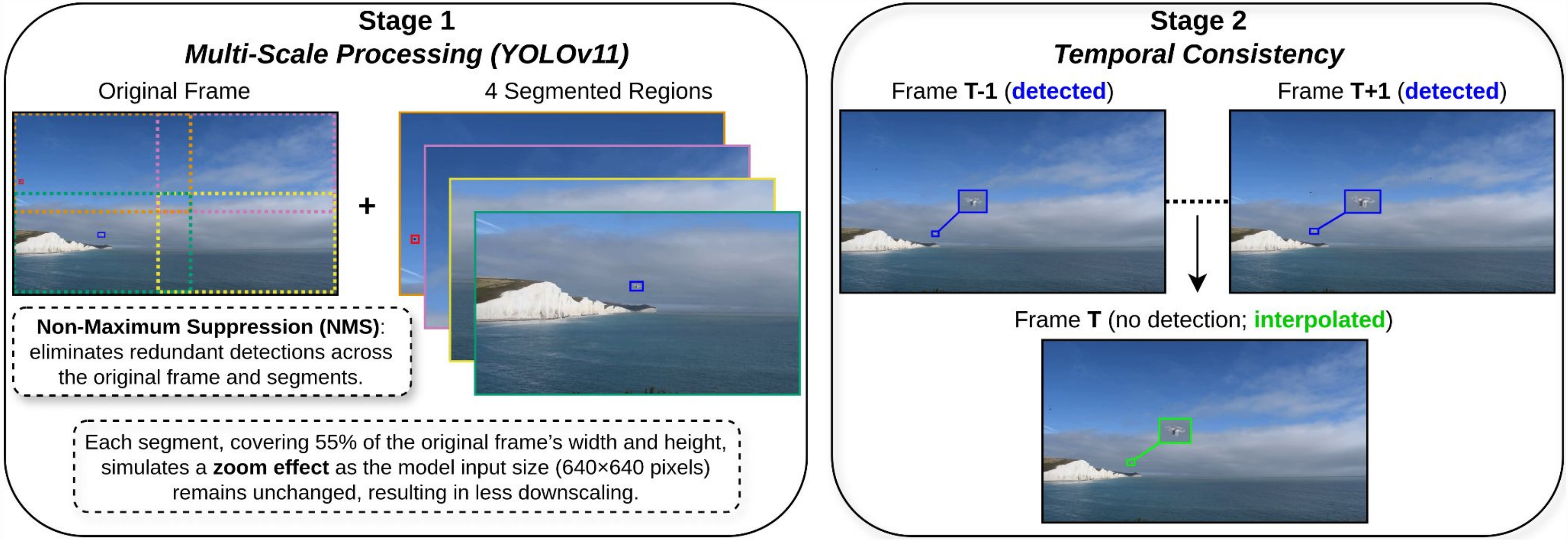
INTERNATIONAL JOINT CONFERENCE ON NEURAL NETWORKS
IJCNN2025
30 JUNE - 5 JULY 2025 | ROME, ITALY
INTERNATIONAL NEURAL NETWORK SOCIETY



Drone Detection

Drone Detection

Drone Detection



R. Laroca, M. dos Santos, D. Menotti, "*Improving Small Drone Detection Through Multi-Scale Processing and Data Augmentation*," in International Joint Conference on Neural Networks (IJCNN), pp. 1-8, June 2025.

Drone Detection

Improving Small Drone Detection Through Multi-Scale Processing and Data Augmentation

Rayson Laroca*,†, Marcelo dos Santos†, and David Menotti†
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†Federal University of Paraná, Curitiba, Brazil
*rayson@ppgia.pucpr.br †{msantos, menotti}@inf.ufpr.br

Abstract–Detecting small drones, often indistinguishable from birds, is crucial for modern surveillance. This work introduces a drone detection methodology built upon the medium-sized YOLOv11 object detection model. To enhance its performance on small targets, we implemented a multi-scale approach in which the input image is processed both as a whole and in segmented parts, with subsequent prediction aggregation. We also utilized a copy-paste data augmentation technique to enrich the training dataset with diverse drone and bird examples. Finally, we implemented a post-processing technique that leverages frame-to-frame consistency to mitigate missed detections. The proposed approach attained a top-3 ranking in the 8th WOSDETC Drone-vs-Bird Detection Grand Challenge, held at the 2025 International Joint Conference on Neural Networks (IJCNN), showcasing its capability to detect drones in complex environments effectively.

Fig. 1. The similarity in size and appearance often makes it challenging to differentiate a drone (blue box) from birds (red box), especially at a distance.

I. Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have experienced a surge in popularity across diverse civil sectors [1], [2]. Their autonomy, flexibility, and affordability have driven their widespread adoption in applications such as search and rescue, package delivery, and remote sensing [3]. However, this rapid expansion presents significant security and privacy challenges [4], [5].

The versatility of drones also makes them susceptible to malicious uses, such as smuggling contraband, conducting intrusive surveillance, and executing physical attacks [6], [7]. Additionally, unauthorized drone operations can violate aviation safety regulations, posing direct threats to civilian aircraft and passengers while causing disruptions at airports, including flight delays [8], [9]. As a result, there is an escalating and urgent demand for advanced drone detection systems to address these growing security and privacy concerns [10], [11].

A major challenge in drone detection is accurately distinguishing drones from birds, given their similarities in size and appearance (see Fig. 1). Moreover, practical systems must be capable of identifying drones at long distances to allow for timely responses. This requires detecting very small

seeks to attract research focused on developing novel signal-processing solutions for the problem of distinguishing birds from drones at long distances. To support participants, the consortium provides a video dataset, which is inherently difficult to acquire given the specific conditions and permissions required for drone flights. The dataset is progressively expanded with each challenge installment [13], [14] and subsequently released to the research community.

Considering the preceding discussion, this work presents a drone detection methodology that leverages the state-of-the-art YOLOv11 object detection model [15]. To enhance the detection of small-scale drones, we implemented a multi-scale approach that processes the input image both as a whole and in segmented components (simulating a zoom effect). Additionally, we employed extensive data augmentation, including a copy-paste technique, to increase the representation of drones and birds in the training images. A post-processing stage, utilizing adjacent frame predictions, was also employed to reduce missed detections. Our approach achieved a top-3 ranking (exact position not disclosed at the time of writing) in the 8th

Data Science

Goal

- My goal is to share both theoretic and practical knowledge I've gathered over the years in terms of machine learning and data science

- You will apply data science and machine learning techniques in a real-world project

- We will meet every Tuesday (18:15–23:00)

How the Lectures Will Work

- Mixture between concepts and their application
- It is up to you and your team to use each tool (or not) in the project
- **Important**: the idea is that you develop **autonomy**, a key characteristic for any computer scientist

The Project

- Available in Canvas

The Project – Template

- Source code: you must use the Jupyter notebook template provided in Canvas

- Presentation: details will be provided in Canvas

- Manuscript: you must abide to IEEE’s template, also available in Canvas

Team Formation (via Canvas)

- Form teams of **5 to 6** students – no exceptions;

Team Formation – FAQ

- **Can I work alone?**
  - No, this is a collaborative project requiring teamwork skills.
- **What if I can only find 1-2 people?**
  - You must find additional members or merge with another incomplete team.
- **What if we have 7+ people interested?**
  - Split into multiple teams of 5–6 members each.

Team Formation (via Canvas)

- Form teams of **5 to 6** students - no exceptions;

- **All students must be assigned to a team by next week (March 11)**
    - **Those who are not will be randomly assigned to one☺️**

Keep in Mind

- During lectures and the project, you and your team will have to display **AUTONOMY**;

- If there is something we haven’t talked about, but you think is suitable: Go for it. Experiment. Go beyond.

- It doesn’t matter if you are pursuing an academic or industrial-oriented position, you will have to conduct a research towards your goal and to develop your skills
    - Being able to critically assess data and what they stand for in the real-world is key

Grading

Tabela de composição da nota do RA e da média semestral: o cruzamento entre um item de avaliação e o respectivo peso do item na nota do RA. A nota semestral mínima para a aprovação do estudante na disciplina é 7,0 (sete). A nota máxima possível em uma recuperação é 7,0.

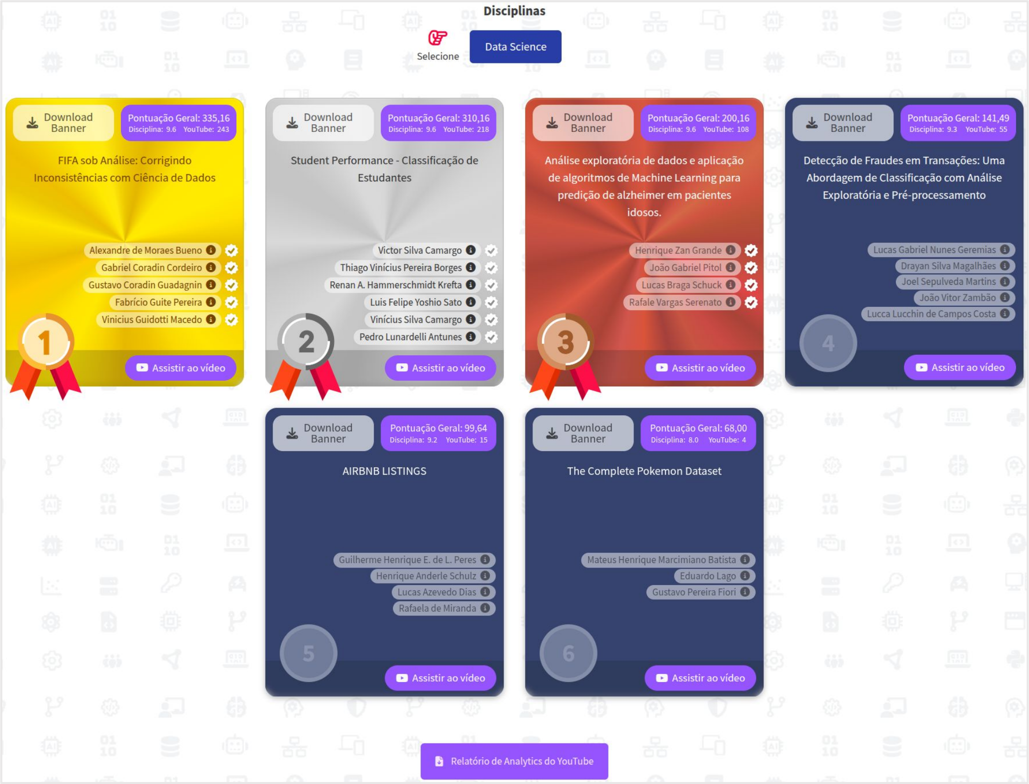
Resultado de Aprendizagem (RA)	Peso no RA						Nota do RA (0 a 10)
	TDE – Checkpoint Código-fonte	TDE – Entrega Final Código-fonte	TDE – Entrega Final Apresentação	TDE – Entrega Final Relatório	Prova 1	Prova 2	
RA1	10%	10%	–	–	40%	40%	Soma ponderada das notas obtidas nas atividades
RA2	20%	20%	20%	40%	–	–	Soma ponderada das notas obtidas nas atividades

Important Dates (subject to change)

Date	Activity
April 8th	Checkpoint submission
April 22th	Exam #1
April 29th	Exam #1 Feedback
May 27th	Exam #2
June 3rd	Exam #2 Feedback + Project submission
June 10th & June 17th	Project presentations

- All dates are available in the course syllabus on Canvas

Galeria de Projetos



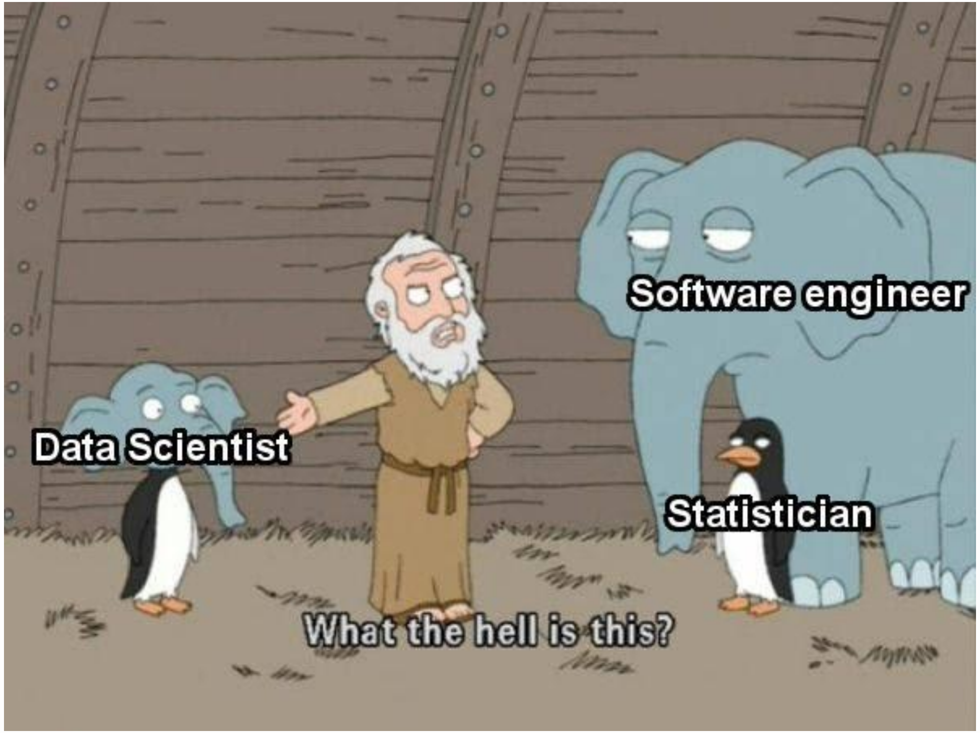
Galeria de Projetos

- All projects with final grading above 80% will be entitled to participate in this process

- Course coordination will provide details in the following weeks

Data Scientist

Data Scientist



Data-Related Professions

- Everybody works with Big Data (or they claim to do so)

- Some roles came up:
    - Data Analyst
    - Data Engineer
    - Data Scientist

- What is the difference between these roles?

Data Analyst

- Analyzes data and uses them to guide decision-making

- Keywords:
  - Data handling
  - Data modelling
  - Reports and visualizations (dashboards) for business needs and decision making

Data Engineer

- Data architecture and preparation

- Keywords:
    - Very tech profile
    - Data pipelining
    - Infrastructure

Data Scientist

- Analyzes and interprets complex data

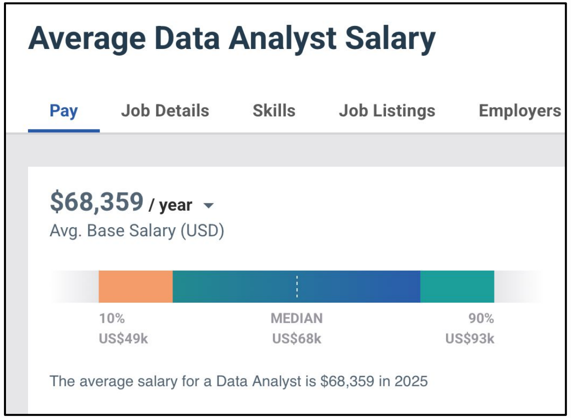
- A mixture between the aforementioned profiles, with a bias towards machine learning (ML)

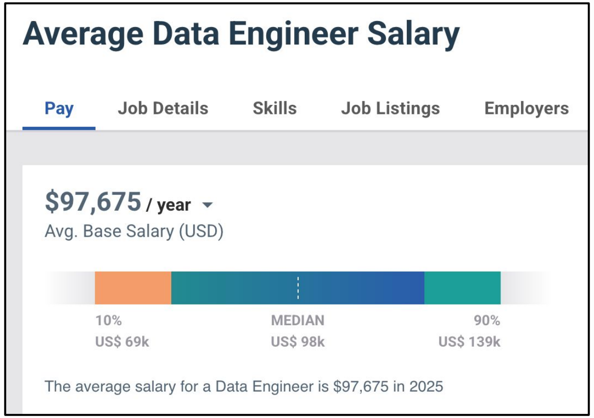
- Keywords:
  - Statistical analysis
  - Deep knowledge in ML
  - Feature engineering and selection, class imbalance, dimensionality reduction, etc.

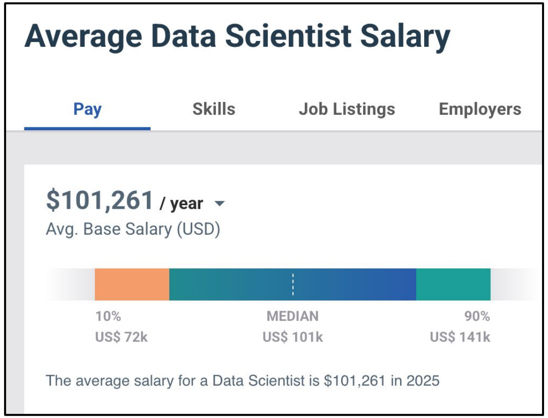
Skills

Data Analyst	Data Engineer	Data Scientist
Report	Programming	Data mining
Effective data visualization	Deep SQL knowledge	Deep ML skills
SQL	Data architecture	Python and/or R skills
Spreadsheet software	Pipelining	Statistical analysis
	ETL	Decision making

Incomes in the USA







Source: https://www.payscale.com/

Tools

Tools

- Python / Jupyter
  - Pandas
  - Scikit-learn
  - Numpy
  - Seaborn
  - Matplotlib
  - Imblearn

- **Note**: if you want to use different tools, feel free to do so!

Google Colab

- https://colab.research.google.com

- Colab has most of the tools we will need

Pandas
Crash Course

Pandas

- Open source library based on **Numpy (C)**
- Data cleaning, preparation and analysis
- Fair performance and productivity
- Allows loading/saving data from/to different sources

Pandas – IO tools

- We can read/write datasets in the following formats:

Format Type	Data Description	Reader	Writer
text	CSV	read_csv	to_csv
text	Fixed-Width Text File	read_fwf	NA
text	JSON	read_json	to_json
text	HTML	read_html	to_html
text	LaTeX	NA	Styler.to_latex
text	XML	read_xml	to_xml
text	Local clipboard	read_clipboard	to_clipboard
binary	MS Excel	read_excel	to_excel
binary	OpenDocument	read_excel	NA
binary	HDF5 Format	read_hdf	to_hdf
binary	Feather Format	read_feather	to_feather
binary	Parquet Format	read_parquet	to_parquet
binary	Apache Iceberg	read_iceberg	to_iceberg
binary	ORC Format	read_orc	to_orc
binary	Stata	read_stata	to_stata
binary	SAS	read_sas	NA
binary	SPSS	read_spss	NA
binary	Python Pickle Format	read_pickle	to_pickle
SQL	SQL	read_sql	to_sql

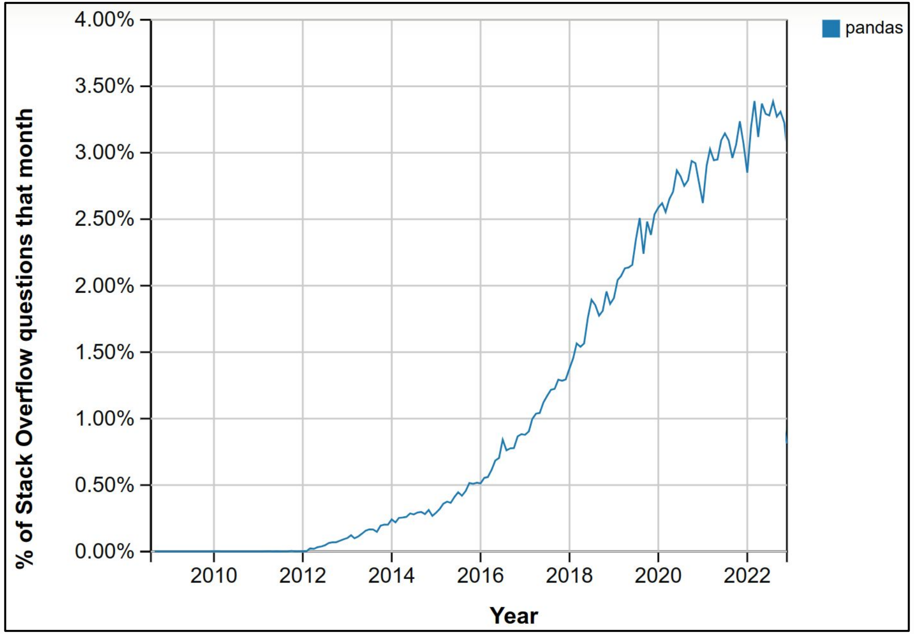
https://pandas.pydata.org/docs/user_guide/io.html

Pandas

- Works with tabular data
- Does a good job when you have data that scales up to millions of rows and hundreds of columns
  - If you need to go further, think about Spark! ☺️
- Used in a multitude of academic and industrial projects

Pandas

- Share of Stack Overflow questions related to pandas over time:



Pandas – DataFrame and Series

- A **Series** is a single column (internally, it is a numpy array)

- A **DataFrame** is a bidimensional structure capable of storing heterogeneous data, i.e., each column (Series) has a different type
  - Therefore, a dataframe is a set of Series

	Name	Team	Number	Position	Age	Height	Weight	College	Salary
0	Avery Bradley	Boston Celtics	0.0	PG	25.0	6-2	180.0	Texas	7730337.0
1	Jae Crowder	Boston Celtics	99.0	SF	25.0	6-6	235.0	Marquette	6796117.0
2	John Holland	Boston Celtics	30.0	SG	27.0	6-5	205.0	Boston University	NaN
3	R.J. Hunter	Boston Celtics	28.0	SG	22.0	6-5	185.0	Georgia State	1148640.0
4	Jonas Jerebko	Boston Celtics	8.0	PF	29.0	6-10	231.0	NaN	5000000.0

Terminology

- Columns are often referred to as **attributes, features, variables, characteristics**;

- Rows are often referred to as **instances, registers, samples, examples.**

	Name	Team	Number	Position	Age	Height	Weight	College	Salary
0	Avery Bradley	Boston Celtics	0.0	PG	25.0	6-2	180.0	Texas	7730337.0
1	Jae Crowder	Boston Celtics	99.0	SF	25.0	6-6	235.0	Marquette	6796117.0
2	John Holland	Boston Celtics	30.0	SG	27.0	6-5	205.0	Boston University	NaN
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4	Jonas Jerebko	Boston Celtics	8.0	PF	29.0	6-10	231.0	NaN	5000000.0

Hint: Loading Data in Google Colab

- The following link has some hints on loading data in Google Colab from Google Drive:
  - https://medium.com/@araujo.dionata/google-colab-importando-arquivos-direto-do-google-drive-5c0fc2798480

Practice

Practice

- Let’s work with pandas and start by analyzing a small dataset

- We will also work with the basic commands of pandas
  - Info, shape, columns, describe, head / tail, column selection, filtering, unique / nunique, group by

df.info()

Number of rows

```
1 df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 397 entries, 0 to 396
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Unnamed: 0     397 non-null    int64
 1   rank           397 non-null    object
 2   discipline     397 non-null    object
 3   yrs.since.phd  397 non-null    int64
 4   yrs.service    397 non-null    int64
 5   sex            397 non-null    object
 6   salary         397 non-null    int64
dtypes: int64(4), object(3)
memory usage: 21.8+ KB
```

Index range

Column names

Column types

RAM usage

Non-missing values per column

df.shape

```
1  df.shape
```

```
(397, 7)
```

This is a tuple!

Number of columns

Number of rows

df.columns

- Useful to check how a column is named exactly

```
1  df.columns
```

```
Index(['Unnamed: 0', 'rank', 'discipline', 'yrs.since.phd', 'yrs.service',
       'sex', 'salary'],
      dtype='object')
```

df.describe()

Numeric columns

```
1  df.describe()
```

Number of values

Standard deviation

Quartiles

	Unnamed: 0	yrs.since.phd	yrs.service	salary
count	397.000000	397.000000	397.000000	397.000000
mean	199.000000	22.314861	17.614610	113706.458438
std	114.748275	12.887003	13.006024	30289.038695
min	1.000000	1.000000	0.000000	57800.000000
25%	100.000000	12.000000	7.000000	91000.000000
50%	199.000000	21.000000	16.000000	107300.000000
75%	298.000000	32.000000	27.000000	134185.000000
max	397.000000	56.000000	60.000000	231545.000000

- Note that non-numeric columns are **NOT** displayed here!

df.head(N) and df.tail(N)

- df.head(N): Returns the **first** *N* rows of a dataset
- df.tail(N): Returns the **last** *N* rows of a dataset

Number of rows

```
1  df.head(10)
```

	Unnamed: 0	rank	discipline	yrs.since.phd	yrs.service	sex	salary
0	1	Prof	B	19	18	Male	139750
1	2	Prof	B	20	16	Male	173200
2	3	AsstProf	B	4	3	Male	79750
3	4	Prof	B	45	39	Male	115000
4	5	Prof	B	40	41	Male	141500
5	6	AssocProf	B	6	6	Male	97000
6	7	Prof	B	30	23	Male	175000
7	8	Prof	B	45	45	Male	147765
8	9	Prof	B	21	20	Male	119250
9	10	Prof	B	18	18	Female	129000

Number of rows

```
1  df.tail(3)
```

	Unnamed: 0	rank	discipline	yrs.since.phd	yrs.service	sex	salary
394	395	Prof	A	42	25	Male	101738
395	396	Prof	A	25	15	Male	95329
396	397	AsstProf	A	8	4	Male	81035

Column Selection

```
df['rank']
```

	rank
0	Prof
1	Prof
2	AsstProf
3	Prof
4	Prof
...	...
392	Prof
393	Prof
394	Prof
395	Prof
396	AsstProf

397 rows × 1 columns

Single column selection

```
df[['rank', 'discipline']]
```

	rank	discipline
0	Prof	B
1	Prof	B
2	AsstProf	B
3	Prof	B
4	Prof	B
...	...	...
392	Prof	A
393	Prof	A
394	Prof	A
395	Prof	A
396	AsstProf	A

397 rows × 2 columns

For multiple columns, we need to pass a **list**

Filtering

```
df_asst = df[df['rank'] == 'AsstProf']
```

Resulting dataframe

Boolean condition
(a.k.a. **mask**)

Filtering

Boolean condition
(a.k.a. **mask**)

```
df_cond = df[(df['rank'] == 'Prof') | (df['salary'] > 110_000)]
```

Resulting dataframe

or

Important: note that each condition is bounded by parentheses!

Logical operators in pandas:

And: &

Or: |

Not: ~

unique and nunique

```
1  df['rank'].unique()
```

```
array(['Prof', 'AsstProf', 'AssocProf'], dtype=object)
```

Returns the unique values in a column

```
1  df['rank'].nunique()
```

```
3
```

Returns the number of unique values

Group by

```
df.groupby(['rank', 'discipline'])['salary'].mean()
```

Columns used for grouping the data

Column(s) selected for statistics calculation

Statistic to be computed

How do we compute several statistics for one or more columns?

Group by

Columns used for grouping the data

Columns selected for statistics calculation

```
df.groupby(['rank', 'discipline']).agg({'salary': ['min', 'max', 'mean'],
                                        'yrs.service': ['min', 'max', 'mean']})
```

Dictionary containing the columns (keys) and lists with statistics to be computed (values)

Statistics to be computed